17 Implement Mobile Price Prediction using appropriate machine learning algorithm

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Create dataset
data = {
    'RAM': [2, 3, 4, 4, 6, 8, 8, 12],
    'Storage': [32, 32, 64, 128, 128, 256, 256, 512],
    'Battery': [3000, 3500, 4000, 4500, 4500, 5000, 5000, 6000],
    'Camera': [8, 12, 16, 16, 32, 48, 64, 108],
    'Price_Range': [0, 0, 1, 1, 2, 2, 3, 3]
}

df = pd.DataFrame(data)

# Input and output
X = df[['RAM', 'Storage', 'Battery', 'Camera']]
y = df['Price_Range']

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# Train model
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

# Predict mobile price range
prediction = model.predict([[8, 256, 5000, 48]])

ranges = {
    0: "Low Price",
    1: "Medium Price",
    2: "High Price",
  3: "Premium Price"
  }

print("Predicted Mobile Price Range:", ranges[prediction[0]])
```

```
# Evaluate model
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Model Accuracy:", round(accuracy * 100, 2), "%")
```

Output:

